CREATIVE CODING · MINI PROJECT

Rested Restless

A light drawing driven by movement

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One word.
Two ways to move.

RESTED

calm, slow, steady

RESTLESS

shaky, fast, nervous

Can we see restlessness as light?

How movement becomes color

Sensor
MPU6050
measures
acceleration

→

One number
XIAO RP2040
"restlessness"

→

Color
RGB LED
blue · green · red

→

Photo
long exposure
light trail

Robot arm UR10e moves the light. Grasshopper plans the path.

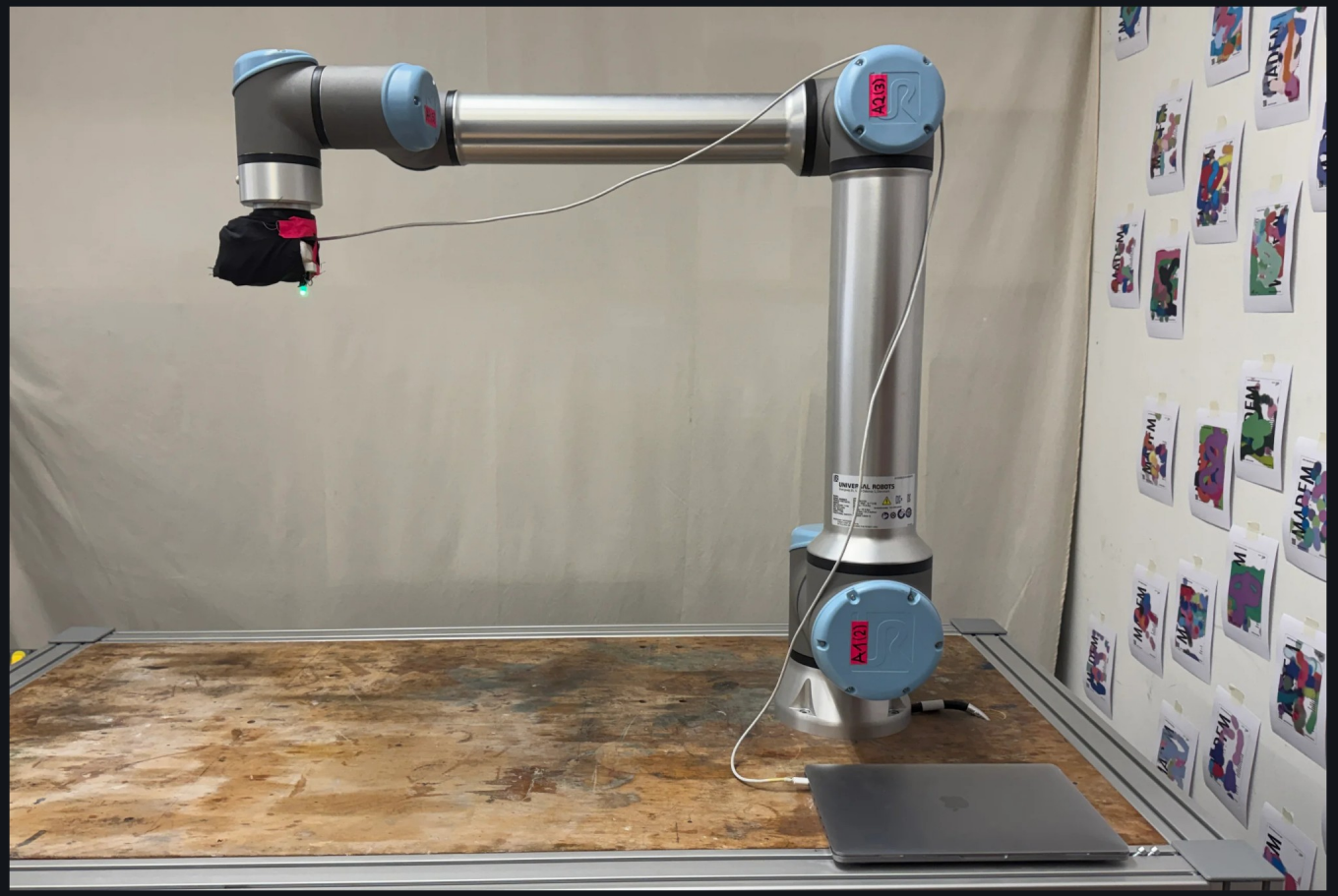
One number decides the color

restlessness = how far the acceleration is from 1 g, smoothed

Blue	below 0.12 · calm
Green	0.12 to 0.30 · in between
Red	above 0.30 · wild

MicroPython · IMU library by Hinch and Plamauer (MIT) · code on GitHub

The setup



Seeed XIAO RP2040

MPU6050 motion sensor

RGB LED

Power from Laptop

UR10e robot arm

The result



RESTED



RESTLESS

What I learned

Sensors measure acceleration, not speed.

Small tuning changes how the result looks.

Robot path planning needs real limits: reach, spacing, blend.

Thank you.

github.com/paulinabul/restlessness